Rev WG-2 (2026-03-22) — ESC Under-Plate Mounting, Grounded Plate Shield, Sensor Consolidation

Changes from layout review and component placement analysis.

Changes

1. **ESC relocated to underside of aluminum mounting plate**

   ESC (80A) now mounts on the **bottom surface** of the 12×24″ aluminum plate, directly above the motor/stern area. This achieves three goals:

   - Aluminum plate acts as EMI shield between ESC (primary noise source) and all sensitive electronics on top surface.
   - Minimizes 3-phase motor lead length (a few inches vs. previous 12″+ run), reducing the highest-EMI antenna in the system.
   - Maintains ~12″ separation from Pixhawk (measured) plus a conductive barrier.

   PWM signal wire from Pixhawk to ESC routed around underside of plate. 50–400 Hz PWM signal has no distance limitation at 2–3 ft on 22 AWG — extend as needed. Route along hull wall, separated from 8 AWG power runs.

   8 AWG power feed to ESC runs from MIDI fuse block through the mounting plate to underside. 3-phase motor leads in dedicated split loom.

   Note: ESC is now the lowest component in the hull — add conformal coat or closed-cell foam splash guard underneath. Ensure small air gap or thermal pad between ESC and plate bottom surface for heat dissipation (plate doubles as heatsink).

2. **Aluminum mounting plate grounded as EMI shield**

   Plate connected to (–) busbar via short 12 AWG jumper with ring terminal. Single ground point only — no second connection to avoid ground loops.

   To prevent short-circuit risk from bare positive terminals contacting the now-grounded plate:

   - **Kapton tape** applied over entire top surface of plate as primary dielectric barrier.

Double-layer under any sharp ring terminal edges or bolt heads.

- **Thermal pads** at converter footprints (12V buck, 5V buck, BEC) provide both thermal coupling and electrical insulation. Confirm thermal pad product is rated as electrically insulating (standard for most thermal pads).
- Kapton + thermal pad combination allows the plate to serve as both EMI shield and heatsink without short-circuit risk.

3. **Sensors consolidated to Pixhawk built-in units**
   - **S7 (IMU) REMOVED** — MPU6050 eliminated from design. Using Pixhawk 4X built-in IMU instead.
   - **S11 (GPS) REMOVED** — NEO6MV2 eliminated from design. Using Pixhawk M8N GPS module (included with Pixhawk 4X).

   Impact on other specs:
   - BOJACK axial ferrite beads reduced from 4 to 2 (S8 pH I2C SDA/SCL only — S7 IMU I2C lines no longer exist).
   - CAT6 bow sensor bundle unchanged (S8, S9, S10 still use all 4 pairs).
   - Copper tape on S7 I2C lines no longer needed.
   - Optional 100nF cap on S11 GPS UART no longer needed.
4. **Copper tape usage updated**

   Still active:
   - P21: 5V buck output → ESP32 (22 AWG pair) — HIGH priority signal cable wrap
   - S1: ESP32-to-Pixhawk UART (22 AWG) — MEDIUM priority signal cable wrap
   - ESC body: Replacing heat shrink tube with copper tape wrap — grounded at one end to (–) busbar. Provides additional shielding on underside of plate.
   - BEC body: Copper tape wrap, grounded at one end
   - Buck converter bodies (12V, 5V): Copper tape wrap, grounded at one end

   Removed from copper tape plan:
   - S7 (IMU I2C) — sensor removed
   - S11 (GPS UART) — sensor removed
   - S10 (TDS analog) — handled by CAT6 shielding in bow sensor bundle

Ground rule unchanged: drain wire grounded at source end only. Single-end grounding is mandatory.

Updated Zone Layout (Bow → Stern)

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Starlink (top deck) → Fuses/Busbars → Converters (top of plate) →
Jetson/Pixhawk/ESP32 (top of plate, aft) → Battery (aft, wide area) →
ESC (UNDERSIDE of plate, stern area) → Motor/Rudder (stern)
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Updated Sensor Designators

ID	Sensor	Interface	Status
S7	~~MPU6050 IMU~~	~~I2C~~	**REMOVED** — using Pixhawk built-in
S8	Atlas EZO-pH	I2C (CAT6)	Active
S9	DS18B20 Temp	1-Wire (CAT6)	Active
S10	TDS Conductivity	Analog (CAT6)	Active
S11	~~NEO6MV2 GPS~~	~~UART~~	**REMOVED** — using Pixhawk M8N
S12	A02YYUW Ultrasonic	UART	Active

Assembly Sequence Update

Between existing steps 6 and 7 (after wiring branch circuits, before installing ferrites):

- Mount ESC on underside of plate with thermal pad. Confirm air gap or pad for heat transfer.
- Route 8 AWG (+) and (–) through plate to ESC.
- Route PWM signal wire from Pixhawk zone around plate edge to ESC underneath.
- Apply conformal coat or foam splash guard to ESC underside.

Before step 1 (plate preparation):

- Apply Kapton tape to entire top surface of aluminum plate.
- Install thermal pads at converter mounting locations.

- Connect 12 AWG ground jumper from plate to (–) busbar.

Parts Impact

Item	Change	Notes
Kapton tape, 2″ wide	**ADD**	~$4, one roll covers plate
12 AWG jumper wire + ring terminals	**ADD**	Short run, plate to (–) busbar
BOJACK ferrite beads	Qty reduced 4 → 2	S8 pH I2C only
MPU6050 breakout	**REMOVED**	Using Pixhawk built-in IMU
NEO6MV2 GPS module	**REMOVED**	Using Pixhawk M8N GPS
Thermal pad (ESC)	**ADD**	For ESC-to-plate thermal coupling underneath